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Systems and methods for providing a robust single carrier radio access network link

Abstract

A network device may determine that a user equipment is entering a first frequency band coverage area provided by the network device and may determine that the user equipment is a power sensitive device or a narrow band device. The network device may receive multiple carrier information associated with the network device based on determining that the user equipment is a power sensitive device or a narrow band device. The network device may process the multiple carrier information, with a machine learning model, to determine a quantity of repetitions to extend uplink coverage for the user equipment and may enable the quantity of repetitions to extend the uplink coverage for the user equipment.

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Background/Summary

BACKGROUND

(1) Multiple carrier aggregation and dual connectivity may improve radio access network (RAN) coverage, RAN link robustness, and capacity for user equipment (UE), such as enhanced mobile broadband (eMBB) devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A-1C are diagrams of an example associated with providing a robust single carrier RAN link for power sensitive or low data rate UEs.
- (2) FIG. 2 is a diagram illustrating an example of training and using a machine learning model.
- (3) FIG. 3 is a diagram of an example environment in which systems and/or methods described herein may be implemented.
- (4) FIG. 4 is a diagram of example components of one or more devices of FIG. 3.
- (5) FIG. 5 is a flowchart of an example process for providing a robust single carrier RAN link for power sensitive or low data rate UEs.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (6) The following detailed description of example implementations refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.
- (7) Multiple carrier aggregation is a technique that increases a data rate per UE, whereby multiple frequency blocks (e.g., called component carriers) are assigned to the same UE. A maximum possible data rate for a UE may be increased as more frequency blocks are assigned to the UE. A sum data rate of a RAN may be increased as well because of better resource utilization. In addition, load balancing is possible with multiple carrier aggregation. Dual connectivity is a feature that

allows UEs to utilize frequencies from different carrier bands, (e.g., both mid-band and millimeter wave (mmWave) frequencies or fourth-generation (4G) and fifth-generation (5G) frequencies) to provide improved RAN coverage and data rate. This is done by implementing multiple carrier aggregation which allows network providers to combine two or more carriers into a single data channel to increase a capacity of the RAN and the data rates. However, some RANs implement a single carrier bandwidth for power sensitive or low data rate UEs (e.g., Internet of Things (IoT) devices), making multiple carrier aggregation and dual connectivity impossible for those UEs. (8) Thus, RANs implementing a single carrier bandwidth may consume computing resources (e.g., processing resources, memory resources, communication resources, and/or the like), networking resources, and/or other resources associated with failing to provide improved RAN coverage and data rates for power sensitive or low data rate UEs, failing to provide improved RAN link robustness for power sensitive or low data rate UEs, failing to provide improved capacity for power sensitive or low data rate UEs, and/or the like.

(9) Some implementations described herein provide a network device (e.g., a RAN device) that provides a robust single carrier RAN link for power sensitive or low data rate UEs. For example, the RAN device may determine that a UE is entering a first frequency band coverage area provided by the RAN device and may determine that the UE is a power sensitive device or a narrow band device. The RAN device may receive multiple carrier information associated with the RAN device based on determining that the UE is a power sensitive device or a narrow band device. The RAN device may process the multiple carrier information, with a machine learning model, to determine a quantity of repetitions to extend uplink coverage for the UE, to determine how to reduce a modulation coding scheme (MCS) to enhance uplink robustness for the UE, and/or to determine how to reduce a block error rate (BLER) to enhance uplink robustness for the UE. The RAN device may enable the quantity of repetitions to extend the uplink coverage for the UE, may reduce the MCS to enhance the uplink robustness for the UE based on determining how to reduce the MCS, and/or may reduce the BLER to enhance the uplink robustness for the UE based on determining how to reduce the BLER.

(10) In this way, the RAN device provides a robust single carrier RAN link for power sensitive or low data rate UEs. For example, the RAN device may include a single carrier robust RAN mechanism that utilizes a machine learning model trained with multiple carrier information associated with the RAN device. The RAN device may trigger a single carrier based on a type of UE and a type of service provided to the UE. For a particular UE (e.g., a power sensitive or low data rate UE, without the ability to utilize multiple carrier aggregation and dual connectivity), the RAN device may utilize the machine learning model to determine a quantity of repetition and slot aggregation with the single carrier to extend uplink (UL) coverage for the particular UE. The RAN device may utilize the machine learning model to determine a reduced MCS matching to enhance uplink robustness for the particular UE. The RAN device may utilize the machine learning model to determine a reduced BLER to enhance uplink robustness for the particular UE. Thus, the RAN device may conserve computing resources, networking resources, and/or other resources that would have otherwise been consumed by failing to provide improved RAN coverage and data rates for power sensitive or low data rate UEs, failing to provide improved RAN link robustness for power sensitive or low data rate UEs, failing to provide improved capacity for power sensitive or low data rate UEs, and/or the like.

(11) FIGS. 1A-1C are diagrams of an example **100** associated with providing a robust single carrier RAN link for power sensitive or low data rate UEs. As shown in FIGS. 1A-1C, example **100** includes a UE **105**, a RAN device **110**, and a core network **115**. Further details of the UE **105**, the RAN device **110**, and the core network **115** are provided elsewhere herein.

(12) Although FIGS. 1A-1C depict the n77, n5, and B5 frequency bands, implementations described herein may be utilized with other frequency bands. For example, an n48 frequency band may be utilized as a New Radio (NR) time division duplexing (TDD) frequency band; an n2

frequency band or an n66 frequency band may be utilized as NR frequency division duplexing (FDD) frequency bands; B13, B66, or B2 frequency bands may be utilized; an n41 frequency band may be utilized, an n71 frequency band may be utilized; and/or the like. Although a low frequency band (e.g., the n5 frequency band or the B5 frequency band) typically provides wider coverage, in a non-collocated deployment other frequency bands may be utilized to achieve coverage and capacity benefits (e.g., a non-collocated B66 frequency band may be utilized as an anchor and may include dual connectivity with the n77 frequency band).

(13) As shown in FIG. 1A, and by reference number **120**, the RAN device **110** may determine that the UE **105** is entering a first frequency band downlink (DL) only zone provided by the RAN device **110**. For example, the RAN device **110** may generate a first frequency band that provides a downlink-only zone. In an example implementation, the first frequency band may include a 3.7 gigahertz (GHz) 5G frequency band (e.g., the n77 frequency band) that is defined by a reference signal received power (RSRP) threshold uplink and an RSRP threshold downlink. When the UE **105** enters (or is approaching) the zone created by the first frequency band, the UE **105** may provide a signal to the RAN device **110**. The RAN device **110** may receive the signal from the UE **105** and may determine that the UE **105** is entering the first frequency band downlink-only zone based on the signal.

(14) As further shown in FIG. 1A, and by reference number **125**, the RAN device **110** may determine whether the UE **105** is a power sensitive device or a narrow band device. For example, when the UE **105** enters the first frequency band coverage area provided by the RAN device **110**, the UE **105** and the RAN device **110** may conduct a registration process for registering the UE **105** with the RAN device **110**. During the registration process, the UE **105** may provide, to the RAN device **110**, capability information that identifies capabilities of the UE **105**, such as whether the UE **105** is a power sensitive device (e.g., an IoT device), whether the UE **105** is not a power sensitive device (e.g., a mobile telephone), whether the UE **105** is a narrow band device, whether the UE **105** is not a narrow band device, and/or the like. In some implementations, the capability information may include frequency band combination information elements indicating whether the UE **105** is capable of multiple carrier aggregation and/or dual connectivity. The RAN device **110** may determine whether the UE **105** is a power sensitive device or a narrow band device based on the capability information. In some implementations, the RAN device **110** may determine that the UE **105** is a power sensitive device or a narrow band device. Alternatively, the RAN device **110** may determine that the UE **105** is not a power sensitive device or a narrow band device.

(15) As shown in FIG. 1B, and by reference number **130**, the RAN device **110** may receive multiple carrier information associated with the RAN device **110** based on determining that the UE **105** is a power sensitive device or a narrow band device. For example, when the RAN device **110** determines that the UE **105** is a power sensitive device or a narrow band device, the RAN device **110** may receive the multiple carrier information associated with the RAN device **110**. The multiple carrier information may include information identifying available component carriers (e.g., frequency blocks), bandwidths of the component carriers, a quantity of component carriers that may be aggregated, and/or the like. In some implementations, the RAN device **110** may monitor the multiple carrier information over time, and may store the multiple carrier information in a data structure (e.g., a database, a table, a list, and/or the like) associated with the RAN device **110**. In such implementations, the RAN device **110** may receive the multiple carrier information from the data structure when the RAN device **110** determines that the UE **105** is a power sensitive device or a narrow band device.

(16) As further shown in FIG. 1B, and by reference number **135**, the RAN device **110** may process the multiple carrier information, with a machine learning model, to determine a quantity of repetitions to extend uplink (UL) coverage for the UE **105** and how to reduce MCS or BLER to enhance UL robustness for the UE **105**. For example, the RAN device **110** may include or have access to a machine learning model. In some implementations, the machine learning model may

include a neural network model, such as an artificial neural network (ANN) model, a convolution neural network (CNN) model, or a recurrent neural network (RNN) model. The RAN device **110** may train the machine learning model, may receive a trained machine learning model, may update the machine learning model, may provide information for updating the machine learning model to another device that updates the machine learning model, and/or the like. Further details of training the machine learning model are described below in connection with FIG. 2.

(17) In some implementations, the RAN device **110** may process the multiple carrier information, with the machine learning model, to determine a quantity of repetitions or slot aggregation to provide so that the RAN device **110** may extend uplink coverage for the UE **105**. As shown in FIG. 1B, the quantity of repetitions or the slot aggregation may extend the uplink coverage (e.g., for the n77 frequency band) from the coverage indicated by the solid line oval to the coverage indicated by the dashed line oval. The extended uplink coverage may encompass the entire n77 downlink-only zone provided by the RAN device **110**.

(18) Alternatively, or additionally, the RAN device **110** may process the multiple carrier information, with the machine learning model, to determine how to reduce MCS so that UL robustness may be enhanced for the UE **105**. MCS defines a quantity of useful bits that can be carried by one symbol (e.g., a resource element) or a quantity of useful bits that can be transmitted per resource element. MCS depends on radio signal quality in a wireless link, where a higher quality radio signal provides a higher MCS (e.g., more useful bits can be transmitted by a symbol) and a lower quality radio signal result in a lower MCS (e.g., less useful bits can be transmitted by a symbol). Thus, the machine learning model may determine how to reduce the MCS for the downlink-only zone so that uplink robustness may be enhanced for the UE **105**.

(19) Alternatively, or additionally, the RAN device **110** may process the multiple carrier information, with the machine learning model, to determine how to reduce BLER so that UL robustness may be enhanced for the UE **105**. BLER is a decoding failure rate of transport blocks (e.g., data blocks). A greater BLER results in more data blocks experiencing decoding failure and a lower BLER results in less data blocks experiencing decoding failure. Thus, the machine learning model may determine how to reduce the BLER for the downlink-only zone so that uplink robustness may be enhanced for the UE **105**.

(20) As shown in FIG. 1C, and by reference number **140**, the RAN device **110** may enable the quantity of repetitions to extend uplink coverage for the UE **105**, may reduce MCS to enhance UL robustness for the UE **105**, and/or may reduce BLER to enhance UL robustness for the UE **105**. For example, the RAN device **110** may enable the quantity of repetitions or the slot aggregation to extend the uplink coverage (e.g., for the n77 frequency band) for the UE **105** from the solid line oval to the dashed line oval. Alternatively, or additionally, the RAN device **110** may reduce the MCS for the downlink-only zone so that uplink robustness may be enhanced for the UE **105**.

Alternatively, or additionally, the RAN device **110** may reduce the BLER for the downlink-only zone so that uplink robustness may be enhanced for the UE **105**. In some implementations, the RAN device **110** may perform various combinations of the aforementioned functions (e.g., enable the quantity of repetitions or the slot aggregation, reduce the MCS, and reduce the BLER).

(21) As further shown in FIG. 1C, and by reference number **145**, the RAN device **110** may enable carrier aggregation with a second frequency band (e.g., an n5 frequency band) and the first frequency band (e.g., the n77 frequency band), for the UE **105**, or may enable dual connectivity, for a third frequency band (e.g., a B5 frequency band) and the first frequency band (e.g., the n77 frequency band), to extend the first frequency band (e.g., the n77 frequency band) coverage for the UE **105** based on determining that the UE **105** is not a power sensitive device or a narrow band device. For example, when the RAN device **110** determines that the UE **105** is not a power sensitive device or a narrow band device, the RAN device **110** may enable, for the UE **105**, carrier aggregation with the second frequency band and the first frequency band. In some implementations, the second frequency band may include an eight hundred and fifty (850)

megahertz (MHz) 5G frequency band.

(22) Alternatively, when the RAN device **110** determines that the UE **105** is not a power sensitive device or a narrow band device, the RAN device **110** may enable dual connectivity, for the third frequency band and the first frequency band, to extend coverage of the first frequency band for the UE **105**. In some implementations, the third frequency band may include an eight hundred and fifty (850) MHz 4G frequency band.

(23) In this way, the RAN device **110** provides a robust single carrier RAN link for power sensitive or low data rate UEs **105**. For example, the RAN device **110** may include a single carrier robust RAN mechanism that utilizes a machine learning model trained with multiple carrier information associated with the RAN device **110**. The RAN device **110** may trigger a single carrier based on a type of UE **105** and a type of service provided to the UE **105**. For a particular UE **105** (e.g., a power sensitive or low data rate UE **105**, without the ability to utilize multiple carrier aggregation and dual connectivity), the RAN device **110** may utilize the machine learning model to determine a quantity of repetition and slot aggregation with the single carrier to extend uplink coverage for the particular UE **105**. The RAN device **110** may utilize the machine learning model to determine a reduced MCS matching to enhance uplink robustness for the particular UE **105**. The RAN device **110** may utilize the machine learning model to determine a reduced BLER to enhance uplink robustness for the particular UE **105**. Thus, the RAN device **110** may conserve computing resources, networking resources, and/or other resources that would have otherwise been consumed by failing to provide improved RAN coverage and data rates for power sensitive or low data rate UEs **105**, failing to provide improved RAN link robustness for power sensitive or low data rate UEs **105**, failing to provide improved capacity for power sensitive or low data rate UEs **105**, and/or the like.

(24) As indicated above, FIGS. **1A-1C** are provided as an example. Other examples may differ from what is described with regard to FIGS. **1A-1C**. The number and arrangement of devices shown in FIGS. **1A-1C** are provided as an example. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIGS. **1A-1C**. Furthermore, two or more devices shown in FIGS. **1A-1C** may be implemented within a single device, or a single device shown in FIGS. **1A-1C** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) shown in FIGS. **1A-1C** may perform one or more functions described as being performed by another set of devices shown in FIGS. **1A-1C**.

(25) FIG. **2** is a diagram illustrating an example **200** of training and using a machine learning model in connection with systems and methods for providing a robust single carrier RAN link for power sensitive or low data rate UEs. The machine learning model training and usage described herein may be performed using a machine learning system. The machine learning system may include or may be included in a computing device, a server, a cloud computing environment, or the like, such as the RAN device **110** described in more detail elsewhere herein.

(26) As shown by reference number **205**, a machine learning model may be trained using a set of observations. The set of observations may be obtained from training data (e.g., historical data), such as data gathered during one or more processes described herein. In some implementations, the machine learning system may receive the set of observations (e.g., as input) from the RAN device **110**, as described elsewhere herein.

(27) As shown by reference number **210**, the set of observations may include a feature set. The feature set may include a set of variables, and a variable may be referred to as a feature. A specific observation may include a set of variable values (or feature values) corresponding to the set of variables. In some implementations, the machine learning system may determine variables for a set of observations and/or variable values for a specific observation based on input received from the RAN device **110**. For example, the machine learning system may identify a feature set (e.g., one or more features and/or feature values) by extracting the feature set from structured data, by

performing natural language processing to extract the feature set from unstructured data, and/or by receiving input from an operator.

(28) As an example, a feature set for a set of observations may include a first feature of first carrier information, a second feature of second carrier information, a third feature of third carrier information, and so on. As shown, for a first observation, the first feature may have a value of first carrier information 1, the second feature may have a value of second carrier information 1, the third feature may have a value of third carrier information 1, and so on. These features and feature values are provided as examples, and may differ in other examples.

(29) As shown by reference number **215**, the set of observations may be associated with a target variable. The target variable may represent a variable having a numeric value, may represent a variable having a numeric value that falls within a range of values or has some discrete possible values, may represent a variable that is selectable from one of multiple options (e.g., one of multiples classes, classifications, or labels) and/or may represent a variable having a Boolean value. A target variable may be associated with a target variable value, and a target variable value may be specific to an observation. In example **200**, the target variable is a quantity of repetitions, which has a value of quantity of repetitions 1 for the first observation. The feature set and target variable described above are provided as examples, and other examples may differ from what is described above.

(30) The target variable may represent a value that a machine learning model is being trained to predict, and the feature set may represent the variables that are input to a trained machine learning model to predict a value for the target variable. The set of observations may include target variable values so that the machine learning model can be trained to recognize patterns in the feature set that lead to a target variable value. A machine learning model that is trained to predict a target variable value may be referred to as a supervised learning model.

(31) In some implementations, the machine learning model may be trained on a set of observations that do not include a target variable. This may be referred to as an unsupervised learning model. In this case, the machine learning model may learn patterns from the set of observations without labeling or supervision, and may provide output that indicates such patterns, such as by using clustering and/or association to identify related groups of items within the set of observations.

(32) As shown by reference number **220**, the machine learning system may train a machine learning model using the set of observations and using one or more machine learning algorithms, such as a regression algorithm, a decision tree algorithm, a neural network algorithm, a k-nearest neighbor algorithm, a support vector machine algorithm, or the like. After training, the machine learning system may store the machine learning model as a trained machine learning model **225** to be used to analyze new observations.

(33) As shown by reference number **230**, the machine learning system may apply the trained machine learning model **225** to a new observation, such as by receiving a new observation and inputting the new observation to the trained machine learning model **225**. As shown, the new observation may include a first feature of first carrier information X, a second feature of second carrier information Y, a third feature of third carrier information Z, and so on, as an example. The machine learning system may apply the trained machine learning model **225** to the new observation to generate an output (e.g., a result). The type of output may depend on the type of machine learning model and/or the type of machine learning task being performed. For example, the output may include a predicted value of a target variable, such as when supervised learning is employed. Additionally, or alternatively, the output may include information that identifies a cluster to which the new observation belongs and/or information that indicates a degree of similarity between the new observation and one or more other observations, such as when unsupervised learning is employed.

(34) As an example, the trained machine learning model **225** may predict a value of quantity of repetitions A for the target variable of the quantity of repetitions for the new observation, as shown

by reference number **235**. Based on this prediction, the machine learning system may provide a first recommendation, may provide output for determination of a first recommendation, may perform a first automated action, and/or may cause a first automated action to be performed (e.g., by instructing another device to perform the automated action), among other examples.

(35) In some implementations, the trained machine learning model **225** may classify (e.g., cluster) the new observation in a cluster, as shown by reference number **240**. The observations within a cluster may have a threshold degree of similarity. As an example, if the machine learning system classifies the new observation in a first cluster (e.g., a first carrier information cluster), then the machine learning system may provide a first recommendation, such as the first recommendation described above. Additionally, or alternatively, the machine learning system may perform a first automated action and/or may cause a first automated action to be performed (e.g., by instructing another device to perform the automated action) based on classifying the new observation in the first cluster, such as the first automated action described above.

(36) As another example, if the machine learning system were to classify the new observation in a second cluster (e.g., a second carrier information cluster), then the machine learning system may provide a second (e.g., different) recommendation and/or may perform or cause performance of a second (e.g., different) automated action.

(37) In some implementations, the recommendation and/or the automated action associated with the new observation may be based on a target variable value having a particular label (e.g., classification or categorization), may be based on whether a target variable value satisfies one or more threshold (e.g., whether the target variable value is greater than a threshold, is less than a threshold, is equal to a threshold, falls within a range of threshold values, or the like), and/or may be based on a cluster in which the new observation is classified. The recommendations, actions, and clusters described above are provided as examples, and other examples may differ from what is described above.

(38) In some implementations, the trained machine learning model **225** may be re-trained using feedback information. For example, feedback may be provided to the machine learning model. The feedback may be associated with actions performed based on the recommendations provided by the trained machine learning model **225** and/or automated actions performed, or caused, by the trained machine learning model **225**. In other words, the recommendations and/or actions output by the trained machine learning model **225** may be used as inputs to re-train the machine learning model (e.g., a feedback loop may be used to train and/or update the machine learning model).

(39) In this way, the machine learning system may apply a rigorous and automated process to provide a robust single carrier RAN link for power sensitive or low data rate UEs. The machine learning system may enable recognition and/or identification of tens, hundreds, thousands, or millions of features and/or feature values for tens, hundreds, thousands, or millions of observations, thereby increasing accuracy and consistency and reducing delay associated with providing a robust single carrier RAN link for power sensitive or low data rate UEs relative to requiring computing resources to be allocated for tens, hundreds, or thousands of operators to manually provide a robust single carrier RAN link for power sensitive or low data rate UEs using the features or feature values.

(40) As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described in connection with FIG. 2.

(41) FIG. 3 is a diagram of an example environment **300** in which systems and/or methods described herein may be implemented. As shown in FIG. 3, the example environment **300** may include the UE **105**, the RAN device **110**, the core network **115**, and a data network **355**. Devices and/or networks of the example environment **300** may interconnect via wired connections, wireless connections, or a combination of wired and wireless connections.

(42) The UE **105** includes one or more devices capable of receiving, generating, storing, processing, and/or providing information, such as information described herein. For example, the

UE **105** can include a mobile phone (e.g., a smart phone or a radiotelephone), a laptop computer, a tablet computer, a desktop computer, a handheld computer, a gaming device, a wearable communication device (e.g., a smart watch or a pair of smart glasses), a mobile hotspot device, a fixed wireless access device, customer premises equipment, an autonomous vehicle, or a similar type of device.

(43) The RAN device **110** may support, for example, a cellular radio access technology (RAT). The RAN device **110** may include one or more base stations (e.g., base transceiver stations, radio base stations, node Bs, eNodeBs (eNBs), gNodeBs (gNBs), base station subsystems, cellular sites, cellular towers, access points, transmit receive points (TRPs), radio access nodes, macrocell base stations, microcell base stations, picocell base stations, femtocell base stations, or similar types of devices) and other network entities that can support wireless communication for the UE **105**. The RAN device **110** may transfer traffic between the UE **105** (e.g., using a cellular RAT), one or more base stations (e.g., using a wireless interface or a backhaul interface, such as a wired backhaul interface), and/or the core network **115**. The RAN device **110** may provide one or more cells that cover geographic areas.

(44) In some implementations, the RAN device **110** may perform scheduling and/or resource management for the UE **105** covered by the RAN device **110** (e.g., the UE **105** covered by a cell provided by the RAN device **110**). In some implementations, the RAN device **110** may be controlled or coordinated by a network controller, which may perform load balancing, network-level configuration, and/or other operations. The network controller may communicate with the RAN device **110** via a wireless or wireline backhaul. In some implementations, the RAN device **110** may include a network controller, a self-organizing network (SON) module or component, or a similar module or component. In other words, the RAN device **110** may perform network control, scheduling, and/or network management functions (e.g., for uplink, downlink, and/or sidelink communications of the UE **105** covered by the RAN device **110**).

(45) In some implementations, the core network **115** may include an example functional architecture in which systems and/or methods described herein may be implemented. For example, the core network **115** may include an example architecture of a 5G next generation (NG) core network included in a 5G wireless telecommunications system. While the example architecture of the core network **115** shown in FIG. 3 may be an example of a service-based architecture, in some implementations, the core network **115** may be implemented as a reference-point architecture and/or a 4G core network, among other examples.

(46) As shown in FIG. 3, the core network **115** may include a number of functional elements. The functional elements may include, for example, a network slice selection function (NSSF) **305**, a network exposure function (NEF) **310**, an authentication server function (AUSF) **315**, a unified data management (UDM) component **320**, a policy control function (PCF) **325**, an application function (AF) **330**, an access and mobility management function (AMF) **335**, a session management function (SMF) **340**, and/or a user plane function (UPF) **345**. These functional elements may be communicatively connected via a message bus **350**. Each of the functional elements shown in FIG. 3 is implemented on one or more devices associated with a wireless telecommunications system. In some implementations, one or more of the functional elements may be implemented on physical devices, such as an access point, a base station, and/or a gateway. In some implementations, one or more of the functional elements may be implemented on a computing device of a cloud computing environment.

(47) The NSSF **305** includes one or more devices that select network slice instances for the UE **105**. By providing network slicing, the NSSF **305** allows an operator to deploy multiple substantially independent end-to-end networks potentially with the same infrastructure. In some implementations, each slice may be customized for different services.

(48) The NEF **310** includes one or more devices that support exposure of capabilities and/or events in the wireless telecommunications system to help other entities in the wireless telecommunications

system discover network services.

(49) The AUSF **315** includes one or more devices that act as an authentication server and support the process of authenticating the UE **105** in the wireless telecommunications system.

(50) The UDM **320** includes one or more devices that store user data and profiles in the wireless telecommunications system. The UDM **320** may be used for fixed access and/or mobile access in the core network **115**.

(51) The PCF **325** includes one or more devices that provide a policy framework that incorporates network slicing, roaming, packet processing, and/or mobility management, among other examples.

(52) The AF **330** includes one or more devices that support application influence on traffic routing, access to the NEF **310**, and/or policy control, among other examples.

(53) The AMF **335** includes one or more devices that act as a termination point for non-access stratum (NAS) signaling and/or mobility management, among other examples.

(54) The SMF **340** includes one or more devices that support the establishment, modification, and release of communication sessions in the wireless telecommunications system. For example, the SMF **340** may configure traffic steering policies at the UPF **345** and/or may enforce user equipment IP address allocation and policies, among other examples.

(55) The UPF **345** includes one or more devices that serve as an anchor point for intraRAT and/or interRAT mobility. The UPF **345** may apply rules to packets, such as rules pertaining to packet routing, traffic reporting, and/or handling user plane QoS, among other examples.

(56) The message bus **350** represents a communication structure for communication among the functional elements. In other words, the message bus **350** may permit communication between two or more functional elements.

(57) The data network **355** includes one or more wired and/or wireless data networks. For example, the data network **355** may include an IP Multimedia Subsystem (IMS), a public land mobile network (PLMN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a private network such as a corporate intranet, an ad hoc network, the Internet, a fiber optic-based network, a cloud computing network, a third-party services network, an operator services network, and/or a combination of these or other types of networks.

(58) The number and arrangement of devices and networks shown in FIG. **3** are provided as an example. In practice, there may be additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than those shown in FIG. **3**. Furthermore, two or more devices shown in FIG. **3** may be implemented within a single device, or a single device shown in FIG. **3** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of the example environment **300** may perform one or more functions described as being performed by another set of devices of the example environment **300**.

(59) FIG. **4** is a diagram of example components of a device **400**, which may correspond to the UE **105**, the RAN device **110**, the NSSF **305**, the NEF **310**, the AUSF **315**, the UDM **320**, the PCF **325**, the AF **330**, the AMF **335**, the SMF **340**, and/or the UPF **345**. In some implementations, the UE **105**, the RAN device **110**, the NSSF **305**, the NEF **310**, the AUSF **315**, the UDM **320**, the PCF **325**, the AF **330**, the AMF **335**, the SMF **340**, and/or the UPF **345** may include one or more devices **400** and/or one or more components of the device **400**. As shown in FIG. **4**, the device **400** may include a bus **410**, a processor **420**, a memory **430**, an input component **440**, an output component **450**, and a communication component **460**.

(60) The bus **410** includes one or more components that enable wired and/or wireless communication among the components of the device **400**. The bus **410** may couple together two or more components of FIG. **4**, such as via operative coupling, communicative coupling, electronic coupling, and/or electric coupling. The processor **420** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of

processing component. The processor **420** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, the processor **420** includes one or more processors capable of being programmed to perform one or more operations or processes described elsewhere herein.

(61) The memory **430** includes volatile and/or nonvolatile memory. For example, the memory **430** may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory). The memory **430** may include internal memory (e.g., RAM, ROM, or a hard disk drive) and/or removable memory (e.g., removable via a universal serial bus connection). The memory **430** may be a non-transitory computer-readable medium. Memory **430** stores information, instructions, and/or software (e.g., one or more software applications) related to the operation of the device **400**. In some implementations, the memory **430** includes one or more memories that are coupled to one or more processors (e.g., the processor **420**), such as via the bus **410**.

(62) The input component **440** enables the device **400** to receive input, such as user input and/or sensed input. For example, the input component **440** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system sensor, an accelerometer, a gyroscope, and/or an actuator. The output component **450** enables the device **400** to provide output, such as via a display, a speaker, and/or a light-emitting diode. The communication component **460** enables the device **400** to communicate with other devices via a wired connection and/or a wireless connection. For example, the communication component **460** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

(63) The device **400** may perform one or more operations or processes described herein. For example, a non-transitory computer-readable medium (e.g., the memory **430**) may store a set of instructions (e.g., one or more instructions or code) for execution by the processor **420**. The processor **420** may execute the set of instructions to perform one or more operations or processes described herein. In some implementations, execution of the set of instructions, by one or more processors **420**, causes the one or more processors **420** and/or the device **400** to perform one or more operations or processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more operations or processes described herein. Additionally, or alternatively, the processor **420** may be configured to perform one or more operations or processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(64) The number and arrangement of components shown in FIG. 4 are provided as an example. The device **400** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. 4. Additionally, or alternatively, a set of components (e.g., one or more components) of the device **400** may perform one or more functions described as being performed by another set of components of the device **400**.

(65) FIG. 5 is a flowchart of an example process **500** for providing a robust single carrier RAN link for power sensitive or low data rate UEs. In some implementations, one or more process blocks of FIG. 5 may be performed by a network device (e.g., the RAN device **110**). In some implementations, one or more process blocks of FIG. 5 may be performed by another device or a group of devices separate from or including the network device, such as a UE (e.g., the UE **105**). Additionally, or alternatively, one or more process blocks of FIG. 5 may be performed by one or more components of the device **400**, such as the processor **420**, the memory **430**, the input component **440**, the output component **450**, and/or the communication component **460**.

(66) As shown in FIG. 5, process **500** may include determining that a UE is entering a first frequency band coverage area provided by the network device (block **510**). For example, the network device may determine that a UE is entering a first frequency band coverage area provided by the network device, as described above. In some implementations, the first frequency band is a

3.7 gigahertz 5G frequency band. In some implementations, the network device is a RAN device. In some implementations, the first frequency band is defined by an RSRP threshold uplink and an RSRP threshold downlink. In some implementations, the UE is an IoT device.

(67) As further shown in FIG. 5, process **500** may include determining that the UE is a power sensitive device or a narrow band device (block **520**). For example, the network device may determine that the UE is a power sensitive device or a narrow band device, as described above. In some implementations, determining that the UE is a power sensitive device or a narrow band device includes receiving capability information associated with the UE, and determining that the UE is a power sensitive device or a narrow band device based on the capability information.

(68) As further shown in FIG. 5, process **500** may include receiving multiple carrier information associated with the network device (block **530**). For example, the network device may receive multiple carrier information associated with the network device based on determining that the UE is a power sensitive device or a narrow band device, as described above.

(69) As further shown in FIG. 5, process **500** may include processing the multiple carrier information, with a machine learning model, to determine a quantity of repetitions to extend uplink coverage for the UE (block **540**). For example, the network device may process the multiple carrier information, with a machine learning model, to determine a quantity of repetitions to extend uplink coverage for the UE, as described above.

(70) As further shown in FIG. 5, process **500** may include enabling the quantity of repetitions to extend the uplink coverage for the UE (block **550**). For example, the network device may enable the quantity of repetitions to extend the uplink coverage for the UE, as described above.

(71) In some implementations, process **500** includes processing the multiple carrier information, with the machine learning model, to determine how to reduce an MCS to enhance uplink robustness for the UE, and reducing the MCS to enhance the uplink robustness for the UE based on determining how to reduce the MCS. In some implementations, process **500** includes processing the multiple carrier information, with the machine learning model, to determine how to reduce a BLER to enhance uplink robustness for the UE, and reducing the BLER to enhance the uplink robustness for the UE based on determining how to reduce the BLER.

(72) In some implementations, process **500** includes determining that another UE is entering the first frequency band, determining that the other UE is not a power sensitive device or a narrow band device, and enabling carrier aggregation with a second frequency band and the first frequency band, for the other UE, based on determining that the other UE is not a power sensitive device or a narrow band device. In some implementations, the second frequency band is an eight hundred and fifty megahertz 5G frequency band.

(73) In some implementations, process **500** includes determining that another UE is entering the first frequency band, determining that the other UE is not a power sensitive device or a narrow band device, and enabling dual connectivity, with a second frequency band and the first frequency band, to extend coverage of the first frequency band for the other UE based on determining that the other UE is not a power sensitive device or a narrow band device. In some implementations, the second frequency band is an eight hundred and fifty megahertz 4G frequency band.

(74) In some implementations, process **500** includes processing the multiple carrier information, with the machine learning model, to determine a slot aggregation to extend uplink coverage for the UE, and enabling the slot aggregation to extend the uplink coverage for the UE.

(75) Although FIG. 5 shows example blocks of process **500**, in some implementations, process **500** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 5. Additionally, or alternatively, two or more of the blocks of process **500** may be performed in parallel.

(76) As used herein, the term “component” is intended to be broadly construed as hardware, firmware, or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware, firmware, and/or a

combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the implementations. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code—it being understood that software and hardware can be used to implement the systems and/or methods based on the description herein.

(77) As used herein, satisfying a threshold may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

(78) To the extent the aforementioned implementations collect, store, or employ personal information of individuals, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

(79) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiple of the same item.

(80) No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, or a combination of related and unrelated items), and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

(81) In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1. A method, comprising: determining, by a network device, that a user equipment is entering a first frequency band coverage area provided by the network device; determining, by the network device, that the user equipment is a power sensitive device or a narrow band device; receiving, by the

network device, multiple carrier information associated with the network device based on determining that the user equipment is a power sensitive device or a narrow band device; processing, by the network device, the multiple carrier information to determine a quantity of repetitions to extend uplink coverage for the user equipment; and enabling, by the network device, the quantity of repetitions to extend the uplink coverage for the user equipment.

2. The method of claim 1, further comprising: processing, by the network device, the multiple carrier information, with a machine learning model, to determine how to reduce a modulation coding scheme to enhance uplink robustness for the user equipment; and reducing the modulation coding scheme to enhance the uplink robustness for the user equipment based on determining how to reduce the modulation coding scheme.

3. The method of claim 1, further comprising: processing, by the network device, the multiple carrier information, with a machine learning model, to determine how to reduce a block error rate to enhance uplink robustness for the user equipment; and reducing the block error rate to enhance the uplink robustness for the user equipment based on determining how to reduce the block error rate.

4. The method of claim 1, further comprising: determining that another user equipment is entering a first frequency band associated with the first frequency band coverage area; determining that the other user equipment is not a power sensitive device or a narrow band device; and enabling carrier aggregation with a second frequency band and the first frequency band, for the other user equipment, based on determining that the other user equipment is not a power sensitive device or a narrow band device.

5. The method of claim 4, wherein the second frequency band is about an 800-900 megahertz frequency band.

6. The method of claim 1, further comprising: determining that another user equipment is entering a first frequency band associated with the first frequency band coverage area; determining that the other user equipment is not a power sensitive device or a narrow band device; and enabling dual connectivity, with a second frequency band and the first frequency band, to extend coverage of the first frequency band for the other user equipment based on determining that the other user equipment is not a power sensitive device or a narrow band device.

7. The method of claim 6, wherein the second frequency band is about an 800-900 megahertz fourth-generation frequency band.

8. A network device, comprising: one or more processors configured to: determine that a user equipment is entering a first frequency band coverage area provided by the network device; determine that the user equipment is a power sensitive device or a narrow band device; receive multiple carrier information associated with the network device based on determining that the user equipment is a power sensitive device or a narrow band device; process the multiple carrier information to determine a quantity of repetitions to extend uplink coverage for the user equipment and to determine how to reduce a modulation coding scheme to enhance uplink robustness for the user equipment; enable the quantity of repetitions to extend the uplink coverage for the user equipment; and reduce the modulation coding scheme to enhance the uplink robustness for the user equipment based on determining how to reduce the modulation coding scheme.

9. The network device of claim 8, wherein a first frequency band is about a 3300-4200 gigahertz frequency band.

10. The network device of claim 8, wherein the network device is a radio access network device.

11. The network device of claim 8, wherein a first frequency band is defined by a reference signal received power (RSRP) threshold uplink and an RSRP threshold downlink.

12. The network device of claim 8, wherein the user equipment is an Internet of Things device.

13. The network device of claim 8, wherein the one or more processors are further configured to: process the multiple carrier information, with a machine learning model, to determine a slot aggregation to extend uplink coverage for the user equipment; and enable the slot aggregation to

extend the uplink coverage for the user equipment.

14. The network device of claim 8, wherein the one or more processors, to determine that the user equipment is a power sensitive device or a narrow band device, are configured to: receive capability information associated with the user equipment; and determine that the user equipment is a power sensitive device or a narrow band device based on the capability information.

15. A non-transitory computer-readable medium storing a set of instructions, the set of instructions comprising: one or more instructions that, when executed by one or more processors of a network device, cause the network device to: determine that a user equipment is entering a first frequency band coverage area provided by the network device; determine that the user equipment is a power sensitive device or a narrow band device; receive multiple carrier information associated with the network device based on determining that the user equipment is a power sensitive device or a narrow band device; process the multiple carrier information, with a machine learning model, to determine a quantity of repetitions to extend uplink coverage for the user equipment and to determine how to reduce a block error rate to enhance uplink robustness for the user equipment; enable the quantity of repetitions to extend the uplink coverage for the user equipment; and reduce the block error rate to enhance the uplink robustness for the user equipment based on determining how to reduce the block error rate.

16. The non-transitory computer-readable medium of claim 15, wherein the one or more instructions further cause the network device to: process the multiple carrier information, with the machine learning model, to determine how to reduce a modulation coding scheme to enhance uplink robustness for the user equipment; and reduce the modulation coding scheme to enhance the uplink robustness for the user equipment based on determining how to reduce the modulation coding scheme.

17. The non-transitory computer-readable medium of claim 15, wherein the one or more instructions further cause the network device to: determine that another user equipment is entering a first frequency band associated with the first frequency band coverage area; determine that the other user equipment is not a power sensitive device or a narrow band device; and enable carrier aggregation with a second frequency band and the first frequency band, for the other user equipment, based on determining that the other user equipment is not a power sensitive device or a narrow band device.

18. The non-transitory computer-readable medium of claim 17, wherein the second frequency band is an eight hundred and fifty megahertz fifth-generation frequency band.

19. The non-transitory computer-readable medium of claim 15, wherein the one or more instructions further cause the network device to: determine that another user equipment is entering a first frequency band associated with the first frequency band coverage area; determine that the other user equipment is not a power sensitive device or a narrow band device; and enable dual connectivity, with a second frequency band and the first frequency band, to extend coverage of the first frequency band for the other user equipment based on determining that the other user equipment is not a power sensitive device or a narrow band device.

20. The non-transitory computer-readable medium of claim 19, wherein the second frequency band is an eight hundred and fifty megahertz fourth-generation frequency band.
